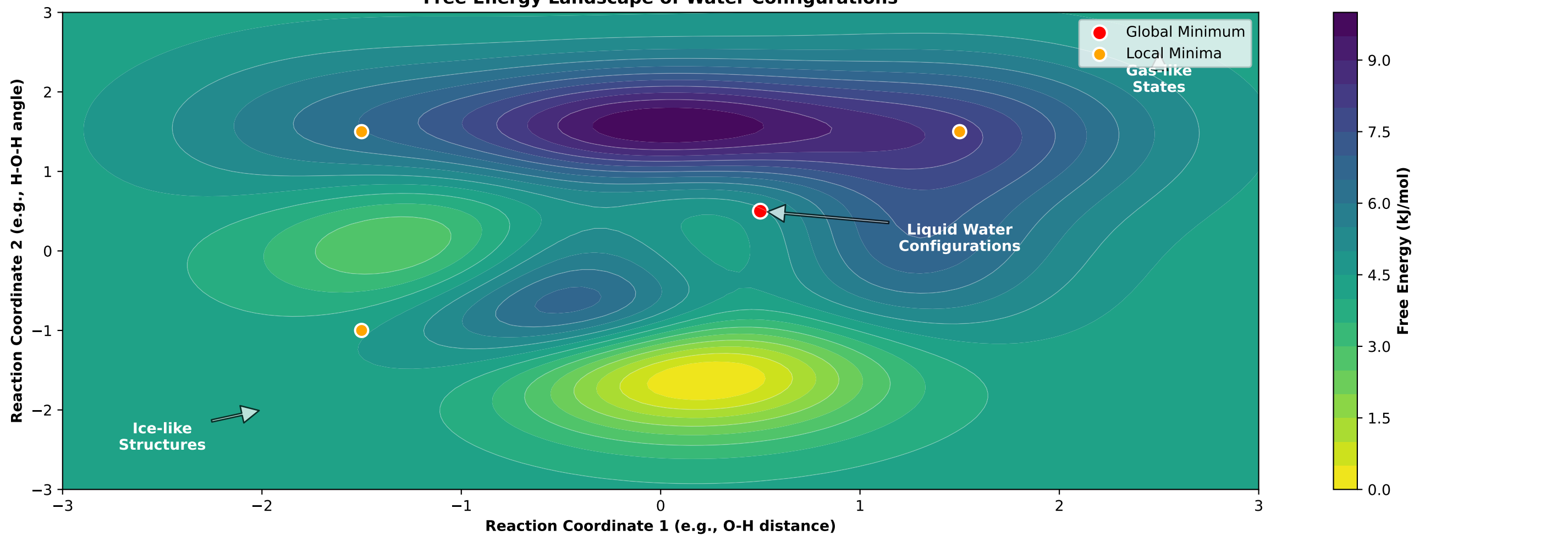
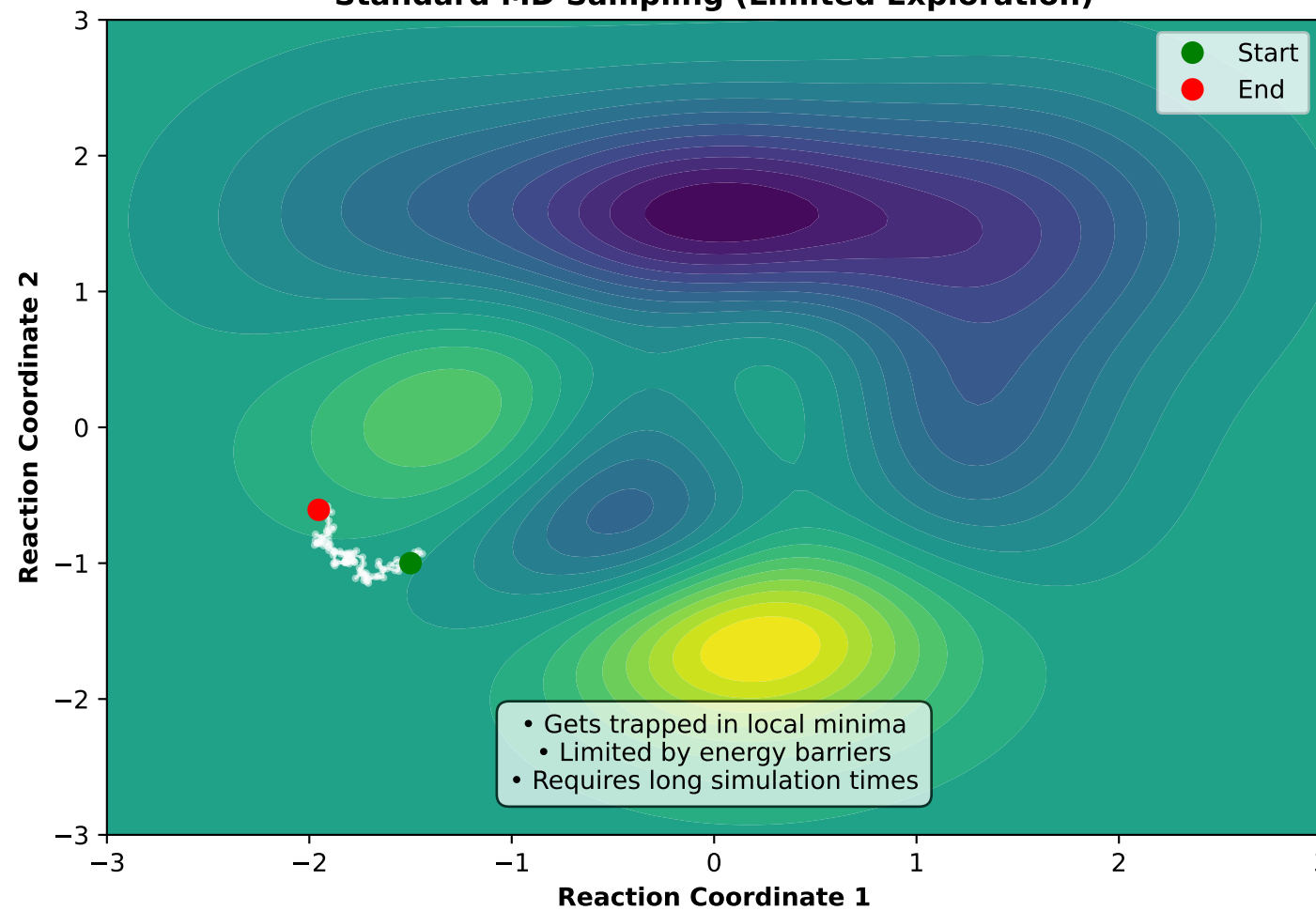


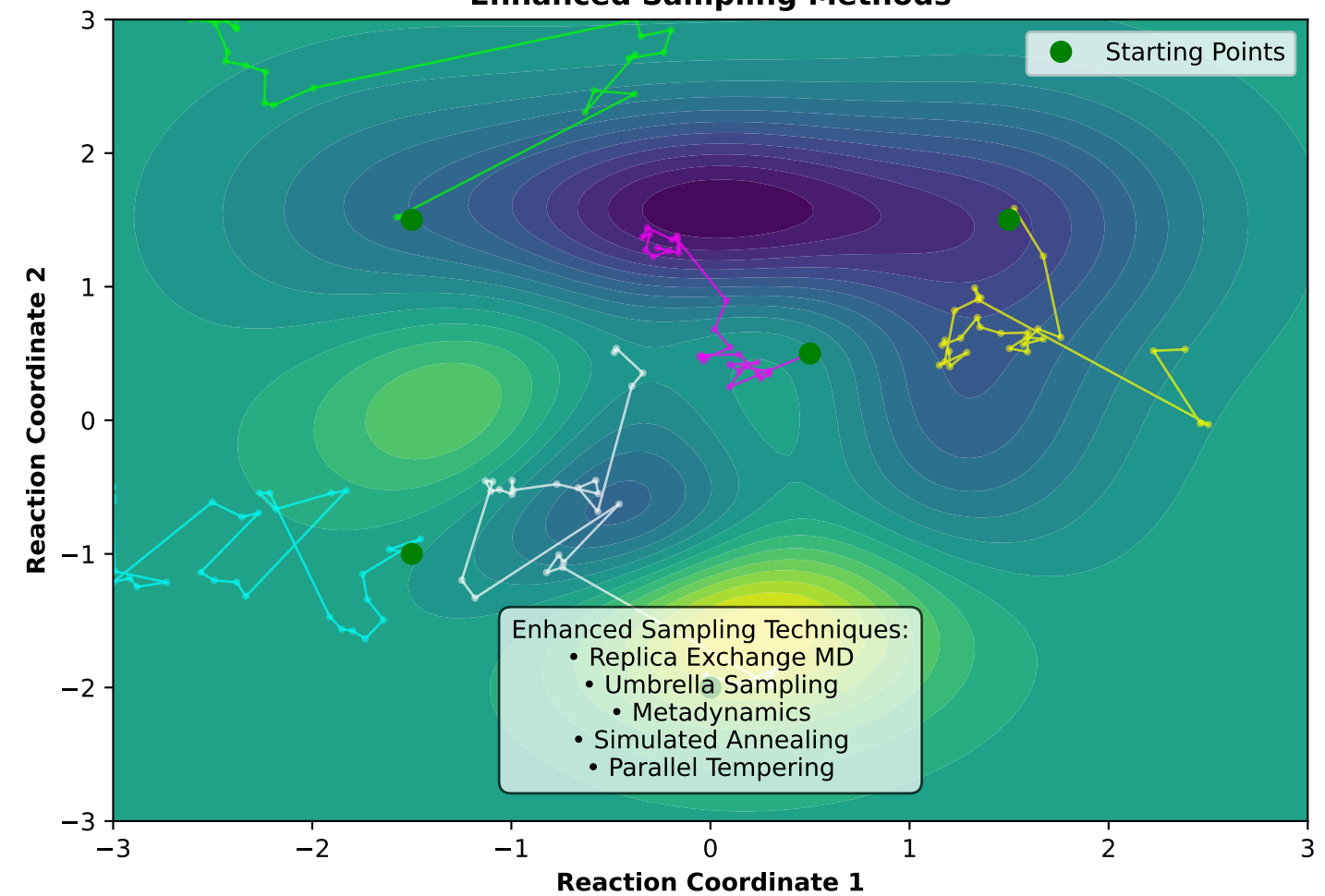
Free Energy Landscape of Water Configurations



Standard MD Sampling (Limited Exploration)



Enhanced Sampling Methods



Sampling Methods in Water Model Simulations:

- Standard MD: Limited by energy barriers, may get trapped in local minima, requires long simulation times
- Enhanced Sampling: Uses various techniques to overcome energy barriers and explore the free energy landscape more efficiently
- For uncorrelated samples: Multiple short simulations from different starting points can be more efficient than a single long one